Evaluation of Impedance and Optimal LQR Control Strategy for Pediatric Gait Exoskeleton Test Benches

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***Abstract*—This paper presents the design and preliminary experimental validation of control strategies intended for a future lower-limb gait exoskeleton. A one-degree-of-freedom (1-DoF) test bench is used to validate a compliant direct impedance control scheme designed to handle external interaction forces. A second test bench, consisting of a two-degree-of-freedom (2-DoF) robotic leg with hip and knee joints, is employed to evaluate trajectory tracking using a state-space Linear Quadratic Regulator (LQR) assisted by a Luenberger observer. The 2-DoF platform is experimentally evaluated using a data acquisition card interfaced with MATLAB/Simulink. Experimental results demonstrate up to a 61% reduction in maximum absolute tracking error with the LQR controller compared to a baseline state-space pole placement approach. These two foundational test-bench stages establish the theoretical and embedded-control framework necessary for the future development of a unified pediatric gait exoskeleton.**

***Index Terms*—Human-Robot Interaction, Optimal Control, LQR, Impedance Control, Pediatric Exoskeleton, Lower-Limb Rehabilitation.**

I. Introduction

The development of lower-limb exoskeletons for pediatric gait rehabilitation represents a complex mechatronic challenge requiring precise dynamic modeling and robust control algorithms [1], [2]. Pediatric rehabilitation demands exceptionally strict safety margins and adaptive systems capable of responding to the growth, anthropometric variability, and varying levels of voluntary motor effort among different pediatric patients [3]–[5]. Accommodating diverse patient profiles without hardware redesign requires real-time algorithmic compliance to maintain safety during involuntary spasticity or active user participation [6], [7].

Historically, rigid position control strategies have been utilized in rehabilitation robotics; however, these approaches fall short in dynamic human-robot interaction (HRI) as they enforce rigid movement patterns and ignore user exertion [8]–[10]. To address this, the paradigm of impedance control, formalized by Hogan [11]–[13], models the robotic joint as a virtual mass-spring-damper system, governing interaction forces and ensuring physical compliance [14]–[16]. Concurrently, optimal control approaches such as the Linear Quadratic Regulator (LQR) minimize energy consumption and tracking errors in multi-joint structures [17]–[20].

To establish the hardware and algorithmic baseline for a future hybrid controller, this paper presents a progressive pedagogical methodology evaluating these control paradigms on preliminary test benches, without human participants. First, a 1-DoF benchmark is used to identify how direct impedance responds to simulated external interactions under static and sinusoidal reference profiles. Second, a 2-DoF robotic leg (hip and knee) evaluates optimal state-space trajectory tracking under coupled non-linear dynamics.

The remainder of this paper is organized as follows. Section II details the progressive methodology, dynamic modeling, and controller formulations. Section III presents the hardware architecture and experimental setups. Section IV evaluates the experimental results across both platforms. Section V discusses scalability toward higher-DoF systems, and Section VI concludes the paper.

II. Methodology and System Modeling

Developing pediatric lower-limb exoskeletons requires balancing compliant physical interaction with trajectory tracking accuracy. To systematically address these conflicting requirements, a two-stage test-bench methodology was established:

1) **1-DoF Compliance Benchmark:** Isolated single-joint physical interaction under direct impedance control. Evaluates how the system yields to external forces.
2) **2-DoF Multi-Joint Tracking:** Scaled to hip and knee joints to evaluate optimal trajectory tracking utilizing an augmented LQR and a Luenberger observer.

A. Part 1: 1-DoF Direct Impedance Control

Fundamentally, impedance control departs from classical trajectory tracking paradigms. Rather than aggressively rejecting environmental disturbances to rigidly maintain a position coordinate, it regulates the dynamic relationship between the exoskeleton and the user [11]. While a strictly rigid controller

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imposes motion regardless of the human's volitional effort, impedance control enforces a programmable mechanical compliance by establishing a target dynamic behavior between the kinematic tracking error and the generalized interaction forces. Defining the task-space error as $\boldsymbol{e}_x(t) = \boldsymbol{x}_d(t) - \boldsymbol{x}(t)$, the virtual restoring force $\boldsymbol{F}_v$ is calculated as:

$$\boldsymbol{F}_v = \boldsymbol{M}_d\ddot{\boldsymbol{e}}_x(t) + \boldsymbol{B}_d\dot{\boldsymbol{e}}_x(t) + \boldsymbol{K}_d\boldsymbol{e}_x(t) \tag{1}$$

where $\boldsymbol{M}_d$, $\boldsymbol{B}_d$, and $\boldsymbol{K}_d$ are the desired positive-definite virtual mass, damping, and stiffness matrices, respectively. To impose this physical behavior on the mechanical structure, $\boldsymbol{F}_v$ is projected into the joint space as a reference torque $\boldsymbol{\tau}_{ref}$ via the transposed analytical Jacobian $\boldsymbol{J}(\boldsymbol{q})$:

$$\boldsymbol{\tau}_{ref} = \boldsymbol{J}^T(\boldsymbol{q})\boldsymbol{F}_v \tag{2}$$

For the current 1-DoF rotational benchmark, this mapping reduces directly to the joint space. Defining the angular tracking error as $e_\theta(t) = \theta_d(t) - \theta(t)$, the reference torque simplifies to $\tau_{ref} = I_d\ddot{e}_\theta(t) + B_d\dot{e}_\theta(t) + K_d e_\theta(t)$. For the experimental implementation, the virtual parameters were set to $I_d = 0.05$ kg $\cdot$ m^2, $B_d = 1.2$ N $\cdot$ m $\cdot$ s/rad, and $K_d = 15$ N $\cdot$ m/rad.

To physically realize this virtual compliance, an inner, high-bandwidth torque-tracking loop is cascaded. The reference motor current is derived using the motor torque constant ($i_{ref} = \tau_{ref}/K_t$), and the armature voltage $V_a(t)$ is regulated via a Proportional-Integral (PI) current controller executed in the embedded hardware:

$$V_a(t) = K_{p,i}\big(i_{ref}(t) - i_m(t)\big) + K_{i,i}\int_0^t \big(i_{ref}(\sigma) - i_m(\sigma)\big)d\sigma \tag{3}$$

This cascaded architecture ensures that when subjected to external user forces, the actuator actively yields according to the prescribed dynamic profile, safely accommodating human interaction instead of blindly forcing a predefined path [14].

B. Part 2: 2-DoF Non-linear Modeling and Linearization

The 2-DoF robotic leg comprises revolute hip and knee joints actuated via 24V DC motors through H-bridges. The coupled dynamic equation is:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) + \boldsymbol{F}(\dot{\boldsymbol{q}}) = \boldsymbol{\tau} \tag{4}$$

where $\boldsymbol{q} = [\theta_1, \theta_2]^T$ contains hip (θ_1) and knee (θ_2) angular coordinates. The kinematic diagram of the 2-DoF link structure is shown in Fig. 1. Parametric identification results are summarized in Table I.

TABLE I
IDENTIFIED PHYSICAL PARAMETERS OF THE JOINT SUBSYSTEM

Parameter	Symbol	Value
Moment of Inertia	J	0.0020 kg $\cdot$ m^2
Viscous Friction Coeff.	β	0.02014 N $\cdot$ m $\cdot$ s/rad
Coulomb Friction Torque	F_c	0.1350 N $\cdot$ m
Motor Torque Constant	K_t	0.50054 N $\cdot$ m/A
Segment Mass	m_2	0.2818 kg
Segment Length	L_2	0.1950 m

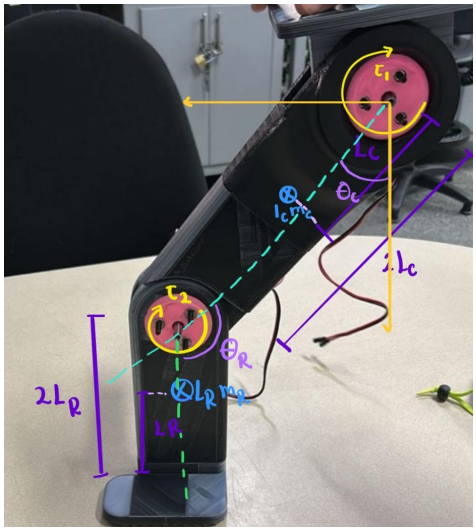


Fig. 1. Kinematic and dynamic model of the 2-DoF gait exoskeleton showing joint angles and link parameters.

Linearizing around vertical stance ($\boldsymbol{q}_0 = [0,0]^T$ rad, $\dot{\boldsymbol{q}}_0 = [0,0]^T$ rad/s) yields the small-signal state-space representation $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}$, and $\boldsymbol{y} = \boldsymbol{C}\boldsymbol{x}$, with state vector $\boldsymbol{x} = [\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2]^T$ and input voltage $\boldsymbol{u} = [V_1, V_2]^T$:

$$\boldsymbol{A} = \begin{bmatrix} 0 & 0 & 1.0000 & 0 \\ 0 & 0 & 0 & 1.0000 \\ -31.5101 & -1.5201 & -3.3295 & 0.1970 \\ -0.3694 & -3.7498 & 0.3753 & -4.0230 \end{bmatrix},$$

$$\boldsymbol{B} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 2.6661 & -0.1876 \\ -0.3005 & 3.8314 \end{bmatrix}, \quad \boldsymbol{C} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

C. Optimal LQR Controller and State Observer Synthesis

1) Augmented LQR Control Strategy: To eliminate steady-state tracking errors induced by joint Coulomb friction and gravitational loads, two integral error states $\boldsymbol{x}_i(t) = \int_0^t (\boldsymbol{y}(\tau) - \boldsymbol{r}(\tau))\,d\tau$ are added, forming the augmented state vector $\boldsymbol{x}_a(t) = [\boldsymbol{x}(t)^T \quad \boldsymbol{x}_i(t)^T]^T$. The optimal state-feedback law minimizes the infinite-horizon cost functional:

$$J = \int_0^\infty (\boldsymbol{x}_a^T\boldsymbol{Q}\boldsymbol{x}_a + \boldsymbol{u}^T\boldsymbol{R}\boldsymbol{u})\,dt \tag{5}$$

Tuning matrices were set to $\boldsymbol{Q} = \text{diag}(1, 1, 80, 80, 4000, 4000)$ and $\boldsymbol{R} = \text{diag}(2, 6)$. Solving the continuous Algebraic Riccati Equation (CARE) yields gain matrix $\boldsymbol{K}_a = [\boldsymbol{K}_x \quad \boldsymbol{K}_i]$.

2) Luenberger State Observer Design: Because angular velocities are not directly measured, a full-order Luenberger observer estimates state vector $\hat{\boldsymbol{x}}(t)$:

$$\dot{\hat{\boldsymbol{x}}} = \boldsymbol{A}\hat{\boldsymbol{x}} + \boldsymbol{B}\boldsymbol{u} + \boldsymbol{L}(\boldsymbol{y} - \boldsymbol{C}\hat{\boldsymbol{x}}) \tag{6}$$

Observer gains $\boldsymbol{L}$ were calculated via pole placement at $p_o = \{-7.88 \pm 5.41i,\ -4.74,\ -163.03\}$. The complete

control structure is depicted in Fig. 2. The optimal control law is expressed as:

$$\boldsymbol{u}(t) = -\boldsymbol{K}_x \hat{\boldsymbol{x}}(t) - \boldsymbol{K}_i \boldsymbol{x}_i(t) = -\boldsymbol{K}_a \boldsymbol{x}_a(t) \tag{7}$$

Although explicitly formulated to minimize the quadratic cost functional J, the augmented LQR framework inherently embeds an impedance-like behavior within the joint space. By partitioning the optimal state-feedback gain matrix into proportional and derivative components, $\boldsymbol{K}_x = [\boldsymbol{K}_p \quad \boldsymbol{K}_v]$, the control law generating the joint actuation torque $\boldsymbol{\tau}_{LQR}$ can be represented in terms of the joint tracking error $\boldsymbol{e}_\theta(t) = \boldsymbol{\theta}_{ref}(t) - \boldsymbol{\theta}(t)$:

$$\boldsymbol{\tau}_{LQR} \approx \boldsymbol{K}_p \boldsymbol{e}_\theta(t) + \boldsymbol{K}_v \dot{\boldsymbol{e}}_\theta(t) + \boldsymbol{K}_i \int_0^t \boldsymbol{e}_\theta(\sigma) d\sigma \tag{8}$$

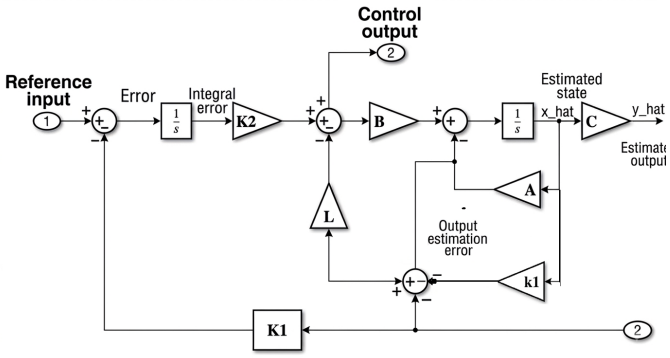


Fig. 2. Augmented state-space LQR control scheme integrated with a Luenberger state observer.

D. Reference Trajectory Generation

1) 1-DoF Reference Profiles: To evaluate parameter tracking and compliance, a static reference and a sinusoidal reference ($\theta_{\text{ref}}(t) = A\sin(2\pi f t)$, $f = 0.5$ Hz) were utilized.

2) 2-DoF Fifth-Order Polynomial Gait Trajectory: For the two-joint exoskeleton, reference trajectories were constructed using 5th-degree polynomial splines with C^2 continuity. These curves were synthesized from manual passive motion recordings obtained by guiding the unpowered joint actuators through target physiological gait positions:

$$\boldsymbol{r}(t) = [\theta_{1,\text{ref}}(t) \quad \theta_{2,\text{ref}}(t)]^T \tag{9}$$

The generated joint reference trajectories are plotted alongside simulated tracking performance in Fig. 3. These trajectories serve as the input for the physical gait validation, establishing the desired kinematic behavior.

III. Experimental Setup and Hardware Implementation

A. 1-DoF Impedance Test Bench

The 1-DoF system comprises a single DC motor, optical encoder, H-Bridge driver, current sensor, and an STM32F446RE microcontroller executing low-level deterministic impedance loops at a sampling rate of 1 kHz.

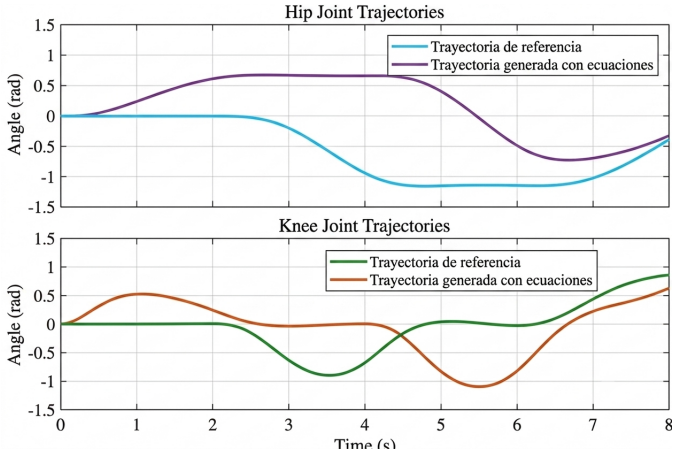


Fig. 3. Synthesized 5th-order polynomial reference gait trajectories and simulated tracking response for hip and knee joints.

B. 2-DoF Experimental Platform

The 2-DoF robotic leg is controlled via a workstation running MATLAB/Simulink connected to a National Instruments NI-USB-6009 Data Acquisition (DAQ) card operating at a 1 kHz sampling rate. This interfaces with the hip and knee 24V DC motors, optical encoders, open-type current sensors, and power drivers. The physical implementation is illustrated in Fig. 4.

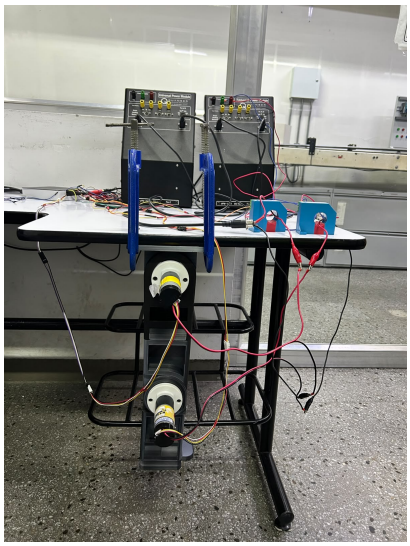


Fig. 4. Physical 2-DoF experimental test bench showing motor actuators, drivers, encoders, and DAQ card setup.

IV. Experimental Results and Comparative Analysis

A. 1-DoF Impedance Response

Retention and tracking tests confirmed that direct impedance control preserves backdrivability. Fig. 5 shows the compliance behavior under external manual displacement force, while

Fig. 6 displays the dynamic response under sinusoidal references. External displacement forces resulted in soft elastic deflection and smooth recovery without dynamic instability. While this serves primarily as a qualitative demonstration of backdrivability, the observed apparent stiffness conforms to the programmed virtual parameter of $K = 15$ N/m.

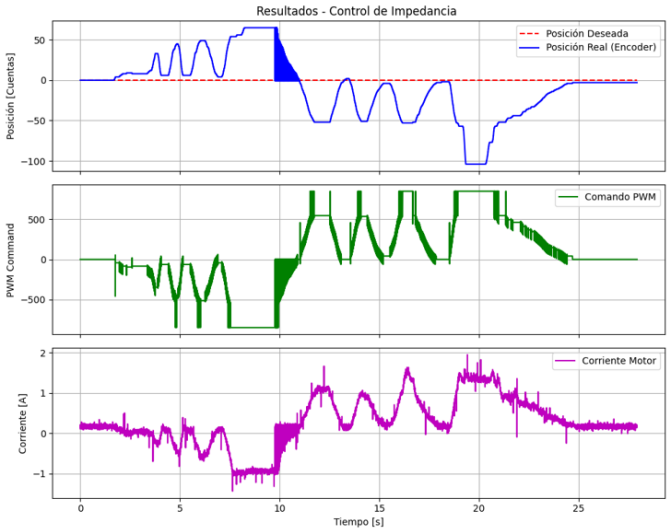


Fig. 5. 1-DoF impedance retention test showing displacement deflection and elastic recovery under external force.

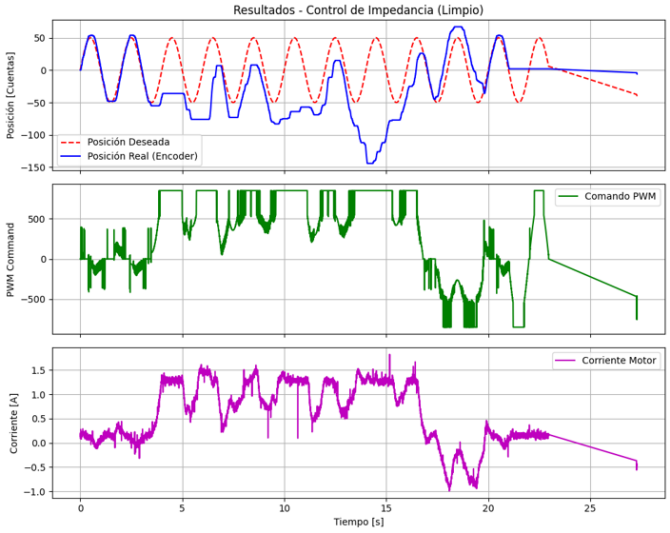


Fig. 6. 1-DoF impedance trajectory tracking response with dynamic parameter adjustment.

B. 2-DoF Trajectory Tracking Analysis

The closed-loop trajectory tracking performance of the 2-DoF robotic leg prototype was experimentally evaluated using the synthesized gait curves. The dynamic response under optimal state-space control is illustrated in Fig. 7.

The augmented LQR scheme demonstrated high reference-following accuracy despite cross-coupling and asymmetric

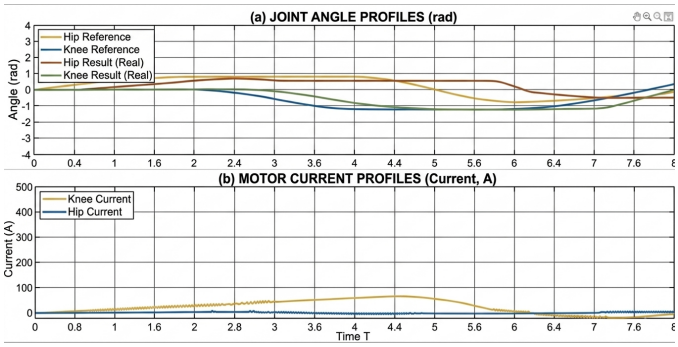


Fig. 7. Experimental trajectory tracking of hip and knee joints using the augmented LQR controller with Luenberger state observer via DAQ interface.

TABLE II
EXPERIMENTAL MAXIMUM ABSOLUTE TRACKING ERROR (2-DOF LEG)

Control Strategy	Hip Error (rad)	Knee Error (rad)
State-Space Pole Placement	± 0.0181	± 0.0230
Optimal LQR Strategy	$\mathbf{\pm 0.0070}$	$\mathbf{\pm 0.0092}$

gravitational loading. As summarized in Table II, the LQR strategy achieved a maximum absolute tracking error of 0.0070 rad for the hip and 0.0092 rad for the knee. This represents an approximate 61% reduction in maximum absolute error compared to the baseline state-space pole placement controller (0.0181 rad and 0.0230 rad, respectively). The integral error states ($\boldsymbol{x}_i$) provided swift torque compensation, suppressing perturbations and restoring exact tracking without exhibiting overshoot.

V. SCALABILITY TOWARD HIGH-DOF SYSTEMS

The evaluation of these distinct test benches establishes a clear technical foundation for scaling control algorithms toward full multi-DoF lower-limb exoskeletons.

- **Dynamic Complexity:** Stage I isolates a single linearized joint to evaluate yield and interaction forces. Stage II incorporates coupled non-linear hip-knee dynamics. Scaling to multi-DoF systems will require extending this formulation to full multi-body dynamics.
- **Control Integration:** For future multi-DoF platforms, the local compliance validated in Stage I and the optimal tracking validated in Stage II will need to be consolidated into a unified hybrid LQR-impedance architecture.
- **Hardware and Processing Scope:** Low-level control transitions from an embedded STM32 microcontroller in Stage I to a high-speed DAQ interface in Stage II. Future full-scale platforms will adopt a distributed embedded network interconnected via high-speed CAN-bus.

VI. CONCLUSION AND FUTURE WORK

This work presented a progressive test-bench methodology investigating two critical control paradigms for pediatric exoskeletons. By isolating the experiments into two distinct platforms, this approach reduces technical risk before building a complete wearable system. The 1-DoF benchmark

demonstrated that direct impedance control provides the backdrivability required for physical interaction, whereas the 2-DoF robotic leg validated an augmented LQR combined with a Luenberger state observer, achieving a 61% reduction in maximum absolute tracking error compared to standard state-space pole placement.

Future work will focus on integrating these two paradigms into a unified hybrid LQR-impedance controller capable of dynamically modulating joint stiffness according to the patient's voluntary effort. The final stage of this research will include embedded implementation on a wearable frame and comprehensive clinical validation under ethics protocols to assess the safety and effectiveness of the platform for pediatric users.

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